MARKETING RESEARCH · SESSION 5

Research Design

The Complete Blueprint

From “we will use a mixed-methods approach” to a blueprint precise enough for another team to execute without ever meeting you.

B L U E P R N T

B L U E P R N T – eight interlocking design decisions

“We will use a mixed-methods approach”

That sentence is not wrong. It is radically incomplete.

It is like an architect telling a client: ‘we will build you a building.’ Technically accurate. Completely uninformative.

Today: build the vocabulary and framework to replace that one sentence with a full design blueprint – precise enough that a different team could execute the research without ever meeting you.

“This is not what I asked for.”

A construction crew receives an architect’s blueprint and starts building. Three weeks in, the client visits the site and says this. Give three possible reasons why – and the stage at which each failure occurred.

1 Wrong problem documented

Failure at the purpose / objectives stage – covered in Sessions 2–3.

2 Right problem, wrong structure

Failure in the design blueprint itself – today’s topic.

3 Right blueprint, poor execution

Failure in fieldwork and data collection – covered in Session 16.

Focus is now entirely on failure #2 – having the right problem but making wrong or incomplete structural decisions before the research starts.

The BLUEPRNT Framework

B	**Background & Research Purpose**	What is this research for? What design type does that require?
L	**Logic of Time**	Snapshot, or track change over time?
U	**Unit of Analysis**	Who or what is being studied – person, household, firm, market?
E	**Evidence Type**	Primary or secondary; qualitative or quantitative?
P	**Procedure for Data Collection**	Survey, interview, observation, experiment, or secondary source?
R	**Reach: Sampling Approach**	Who is included, how many, how selected?
N	**Numbers & Measurement**	What is measured, at what level of precision?
T	**Techniques of Analysis**	How will the data be analysed to answer the objectives?

Two rules for using BLUEPRNT

1 Order matters

Each decision constrains the ones that follow. Purpose (B) determines the time dimension (L); time + unit (U) constrain the data collection procedure (P); sampling (R) must match the procedure; analysis (T) must match data type (E) and measurement (N).

A design built by picking components independently is a design that fails in execution.

2 A framework, not a finish line

This framework helps in asking the right questions. It does not builds the skill of answering every one optimally.

A research design is not a method. It is a set of eight interlocking decisions – get one wrong and the others unravel.

B

BACKGROUND and Research Purpose

The ‘B’ in BLUEPRNT –
Why does this research exists, and
What type of design its purpose requires.

Three fundamental design types

Exploratory	Descriptive	Causal (Experimental)
Problem not yet well-defined. Produces hypotheses & constructs – not generalisable, causal, or precise findings.	Know what to measure; need how much / how often / among whom. Produces precise, generalisable estimates – not causal claims.	Establish that one variable causes change in another. Produces evidence of causation – not rich context or broad generalisability.

***NOT MUTUALLY EXCLUSIVE**:*
a research study often moves through all three in sequence – exploratory to discover, descriptive to measure, causal to test. Skipping exploratory because it "feels unscientific" is the single most common design error.

Design Type	When It's the Right Call	What It Can and Cannot Produce	What It Rules Out
Exploratory	Research problem is not yet well-defined; Researcher needs to understand the territory before committing to specific hypotheses or measurements	Rich insight into the range and nature of the phenomena; hypotheses and constructs that a future study can test rigorously; it cannot produce generalisable, statistically defensible findings on its own	Causal claims; precise prevalence estimates; findings that hold beyond the specific people or contexts studied
Descriptive	The researcher knows what to measure and needs to establish how much, how often, or among whom – precisely and generalising	Statistically precise estimates of prevalence, frequency, or distribution across a defined population; associations between variables (not causation)	Causal claims; explanation of why patterns exist rather than that they exist
Causal (Experimen-tal)	The researcher needs to establish that one variable (the treatment) actually causes a change in another (the outcome), ruling out alternative explanations	Evidence of causation; basis for confident intervention decisions ('if we do X, Y will change by Z')	Rich contextual understanding of why the effect occurs; generalisability beyond the experimental setting without further study

The Insurance Company's Claims Problem

SCENARIO

An insurance company notices a sharp rise in fraudulent motor claims over two quarters. The chief analytics officer wants to commission research.

Three internal stakeholders propose different research purposes:

- the fraud team wants to 'understand how fraudsters think and what tactics they use';
- the actuarial team wants to 'estimate what proportion of claims in each vehicle category are likely fraudulent'; and
- the operations team wants to 'test whether adding a mandatory video call with a claims assessor reduces the fraud rate.'

DISCUSSION PROMPT

Can a single study serve all three objectives?
What design type will best answer the questions?

The Insurance Company's Claims Problem

Fraud team	*"How fraudsters think and what tactics they use"*	**Exploratory**
Actuarial team	*"What proportion of claims per category are fraudulent"*	**Descriptive**
Operations team	*"Does a mandatory video call reduce the fraud rate"*	**Causal**

A single study cannot serve all three – different data, different inferential logic, different outputs. The ideal programme runs them in sequence: exploratory → descriptive → causal.

The EdTech Platform's Dropout Problem

SCENARIO An online learning platform has seen a 35% increase in course dropout rates over two semesters.

- The head of product says: 'We need to understand why students are dropping out – is it the content, the pace, life circumstances, or something else we haven't thought of?'
- The head of data says: 'I want to know the exact dropout rate by course category, instructor rating, and device type.'
- The head of growth says: 'I want to test whether adding a weekly SMS reminder to at-risk students reduces dropout.'

DISCUSSION PROMPT

Classify each objective and explain what would be wrong with using a single design to address all three.

The EdTech Platform's Dropout Problem

Head of Product	*"We need to understand why students are dropping out – is it the content, the pace, life circumstances, or something else we haven't thought of?"*	**Exploratory**
Head of Data	*"I want to know the exact dropout rate by course category, instructor rating, and device type."*	**Descriptive**
Head of Growth	*"I want to test whether adding a weekly SMS reminder to at-risk students reduces dropout."*	**Causal**

L

LOGIC OF TIME

Will this research take a single snapshot, or track how things change over time?

Cross-sectional vs. Longitudinal

Cross-sectional	One point in time – a photograph	*Cannot support causal claims or detect trends*
Longitudinal – Panel	Same units, multiple waves	*Expensive; attrition; panel conditioning risk*
Longitudinal – Cohort	Defined group, may sample different individuals per wave	*Cannot measure individual-level change*
Longitudinal – Trend	Different samples of same population, multiple waves	*Cannot track individual change; sample shifts can mask real trends*

Cross-sectional designs produce associations.
Longitudinal designs CAN – but do not automatically – produce evidence of change.

The Telecom's Customer Satisfaction Mystery

SCENARIO A telecom operator's annual customer satisfaction survey (conducted every December with a fresh random sample of 2,000 subscribers) shows satisfaction declining:

74% in Year 1, 68% in Year 2, 61% in Year 3.

The CMO concludes: 'Our existing customers are becoming less satisfied.'

The chief research officer pushes back, noting the company acquired 40% more subscribers over the three-year period – a large portion from lower-income segments through a budget tariff launched in Year 2.

DISCUSSION PROMPT

What time design is this study using, and why might the CMO's conclusion be wrong? What time design would be needed to support that conclusion?

The Telecom's Customer Satisfaction Mystery

Annual survey, fresh random sample each December: satisfaction 74% → 68% → 61% over 3 years. Company also acquired 40% more subscribers via a Year-2 budget tariff.

CMO: "Our existing customers are becoming less satisfied."

This is a **TREND study** – different individuals sampled each year. The design cannot support a claim about the same customers changing.

More plausible:
compositional change – new lower-income segment customers, not attitude change.

TO SUPPORT THE CMO'S CLAIM

A PANEL design is needed – tracking the same customers across all three years, comparing long-tenure vs. newly-acquired trajectories separately.

Lesson: the time design fundamentally changes what you can conclude – even from the same question, same population size, same period.

The FMCG Brand Tracker

SCENARIO

A consumer goods company runs a quarterly brand health tracker, surveying 500 urban consumers each quarter for six years on brand awareness, consideration, preference, and NPS for its detergent brand and four competitors.

- The marketing director presents the data and says: 'Our brand preference score has been flat for six years while Competitor A has grown 12 points. We need to understand why the same consumers are preferring them more over time.'
- The research director corrects her: 'These are not the same consumers – we sample 500 new people every quarter.'

DISCUSSION PROMPT

What design is this, what can and cannot be concluded from it, and what would be needed to support the marketing director's claim?

U

UNIT of Analysis

*Not the topic – the unit. One of the **most commonly skipped** decisions, with the biggest downstream cost.*

The entity data is collected about – not the respondent

Individual person	**Household / family**	**Organisation / firm**
Product / SKU / brand	**Market / geography**	**Event / transaction**

Named error: getting this wrong is the ECOLOGICAL FALLACY (concluding about individuals from group-level data) or ATOMISTIC FALLACY (concluding about groups from individual-level data).

Consequences cascade into sampling (units are sampled, NOT topics), measurement, and analysis (many statistical techniques assume independence – violated when transactions nest within customers).

The B2B Software Renewal Study

SCENARIO A SaaS company wants to understand what drives enterprise clients to renew or not renew their annual software subscription.

The research team designs a survey targeting 'people at client companies who are involved in the renewal decision.' They send the survey to the primary contact on file for each account – typically the IT manager who manages the technical implementation.

They collect 280 responses from 280 accounts and draw conclusions about 'which accounts are likely to renew and why.'

When the findings are presented to the CRO, she asks: 'Did you speak to the CFO? In most of our enterprise accounts, the CFO makes the final call on renewal, not the IT manager.'

DISCUSSION PROMPT What is the stated unit of analysis, what is the actual unit of analysis in the data collected, and what problem does the mismatch create?

The B2B Software Renewal Study

Survey targets “people involved in the renewal decision” – but is sent only to the IT manager on file. 280 responses, 280 accounts.

CRO: “Did you speak to the CFO? In most enterprise accounts, the CFO makes the final call – not the IT manager.”

STATED unit of analysis

The account (organisation)

The research is explicitly about account-level renewal decisions.

ACTUAL data collected

One IT manager, per account

Not representative of the account-level decision – IT managers care about implementation; CFOs care about ROI and budget fit.

E

EVIDENCE Type

Two orthogonal decisions live here: data source, and data form. Both must be made.

Two decisions, four combinations

*Decision 1: **Source** (primary / secondary) × Decision 2: **Form** (qualitative / quantitative)*

	QUALITATIVE	QUANTITATIVE
PRIMARY	In-depth interviews, focus groups, ethnography, projective techniques	Surveys, structured observations, experiments
SECONDARY	Social posts, reviews, call transcripts, internal reports	CRM/transaction data, web analytics, census, syndicated panels

The largest, most underused asset: "secondary quantitative" – CRM, transaction logs, clickstream – answers questions faster and cheaper than a new survey.

The Retail Bank's Mortgage Drop-Off

SCENARIO A retail bank notices that mortgage application completion rates have dropped from 68% to 51% over 18 months. The head of digital banking wants to commission research.

An analyst proposes 'a large survey of 1,000 customers who started but did not complete an application.'

The head of digital analytics says: 'Before we spend money on a survey, can we look at what we already have?' She pulls the bank's web analytics and finds that 73% of drop-offs occur at the same step: the income documentation upload screen. She also pulls customer support tickets and finds a spike in complaints about the upload interface starting 19 months ago, coinciding with a website redesign.

DISCUSSION What data source and form was the analyst proposing, and what did the analytics head find by using a different evidence type first? What does this tell us about sequencing the Evidence Type decision?

The Retail Bank’s Mortgage Drop-Off

Completion rate fell 68% → 51%. A junior analyst proposes a 1,000-customer survey. The analytics head checks existing data first.

PROPOSED

Primary quantitative

A new survey of 1,000 customers who started but didn’t complete – 3 weeks, real budget.

USED INSTEAD

Secondary quant + qual

Web analytics: 73% of drop-offs occur at the income-upload screen. Support tickets spiked after a redesign 19 months ago – found within hours.

This reframes – not eliminates – the survey: a small, targeted usability study on the upload screen, not a broad open-ended one.

The D2C Food Brand's Product Review Analysis

SCENARIO

A direct-to-consumer healthy snacks brand has received 3,400 customer reviews on its e-commerce platform over two years.

The marketing manager wants to understand 'what customers value most about our products.' She asks the research team to design a customer preference survey.

The research analyst suggests running a text analysis of the existing reviews first, categorising them by sentiment and theme, before deciding whether a survey is needed.

DISCUSSION PROMPT

What evidence type is the marketing manager proposing, and what is the analyst suggesting? What would be gained and lost by each approach?

P

PROCEDURE for Data Collection

Students most often say ‘survey’ and stop. This component is much richer than that.

Five families – and three mismatches to avoid

Surveys	Qualitative interviewing	Observation	Experiment	Secondary data extraction

Survey × Exploratory purpose	*Surveys presuppose you know what to ask – exactly what exploratory research is meant to discover.*
Observation × Household unit	*Natural observation captures individual behaviour, not the household's negotiation process.*
Cross-sectional survey × Causal question	*Surveys can show association, never rule out confounders or establish sequence – that needs an experiment.*

Within each family, the sub-method matters too: CATI vs. CAPI vs. online survey trade off coverage, response rate, cost, and quality. The blueprint specifies the sub-method, not just the family.

Procedure	What It Does	Best Suited For
Surveys (structured self-report)	Collects data by asking respondents to answer standardised questions, either in writing, verbally, or digitally	Primary quantitative data; descriptive and some causal designs; large samples; measuring attitudes, perceptions, stated behaviours, and demographics
Qualitative (IDIs, focus groups)	Collects data through guided conversation – either one-on-one (IDI) or in a group (FGD) – using open-ended prompts rather than fixed questions	Primary qualitative data; exploratory designs; understanding meaning, motivation, and language
Observation	Collects data by watching behaviour rather than asking about it. Can be natural (in the field) or structured (in a controlled setting)	Primary qualitative or quantitative data depending on whether the observation is coded; particularly valuable when what people do differs from what they say they do
Experiment	Collects data by manipulating one or more variables under controlled conditions and observing the effect on an outcome variable	Primary quantitative data; causal designs; requires random assignment of units to conditions
Secondary data	Collects data by accessing and organising existing datasets – internal or external	Secondary data of any form; First procedure evaluated for any research objective

The Luxury Hotel's Guest Experience Study

SCENARIO

A five-star hotel chain wants to understand what aspects of the guest experience most differentiate it from competitors in the eyes of high-value guests (those who spend more than ₹25,000 per night).

- The research team proposes an online survey sent to the hotel's loyalty programme members post-stay.
- The general manager reads the proposal and says: 'Our highest-value guests never respond to email surveys. And I'm not sure a questionnaire can capture what it actually feels like to stay here.'

DISCUSSION PROMPT

What procedure is being proposed, and what is wrong with it for this specific unit of analysis, evidence type, and research purpose?

What alternative procedures should the design consider?

The FMCG Company's 'Usage and Attitude' Study

SCENARIO

An FMCG company wants to conduct a Usage and Attitude (U&A) study for its hair care product range across five Indian cities.

The brand manager asks the research agency to 'do a U&A.'

The agency proposes 1,500 face-to-face interviews using a structured questionnaire, covering category usage frequency, brand awareness and usage, brand imagery, product attribute importance and performance, and purchase drivers.

Budget: ₹32 lakhs; Timeline: eight weeks.

DISCUSSION PROMPT

Is the procedure appropriate for the stated research purpose?

What specific procedure details does this design still need to specify before it is complete?

Common Incorrect Combinations

- Proposing a survey for an exploratory research purpose
- Proposing observation for a study whose unit of analysis is the household
- Proposing a cross-sectional survey for a causal research question

Each procedure family has multiple specific methods. Having consequences on cost, response rate, data quality, & coverage.

- E.g. Within surveys,
  - telephone interview, face-to-face interview, self-administered paper questionnaire, and online surveys

  involves trade-offs in
  - coverage (not everyone has internet; not everyone answers the phone),
  - response rate,
  - data quality (interviewers can clarify but can also introduce bias), and
  - cost.

R

REACH: Sampling Approach

Four questions, in order.
None can be left unspecified in a complete blueprint.

Four questions, answered in order

1	**Target Population**	Who are all the people/units the research generalises about?
2	**Sampling Frame**	What list or access mechanism will respondents actually be drawn from?
3	**Sampling Method**	Probability or non-probability selection?
4	**Sample Size**	How many, and what's the justification?

MOST COMMON DESIGN ERROR IN PRACTICE: **Using non-probability sampling for a descriptive objective, then drawing population-level conclusions.**

The Urban Youth Mental Health Study

SCENARIO A mental health NGO in India wants to understand the prevalence of anxiety symptoms among urban college students aged 18–24 and the factors most associated with higher symptom severity.

They commission a research agency.

The agency recruits 800 respondents by posting an invitation on Instagram and Twitter, asking college students to complete a 15-minute online survey.

800 responses are collected in four days.

The agency reports: '70% of urban Indian college students experience clinically significant anxiety symptoms.'

DISCUSSION Identify the target population, the sampling frame used, the sampling method, and explain why the agency's population-level conclusion is not supported by this design.

The Urban Youth Mental Health Study

Target population	Urban Indian college students, 18–24
Sampling frame	Whoever encountered the social post
Sampling method	Non-probability, self-selected convenience

Why the 70% figure is unreliable:

- Sample size ≠ sample quality – a biased sample of 800,000 is still biased.
- Social media users differ systematically from the full population.
- Anxious students may be more likely to respond – response bias correlated with the outcome.
- No margin of error can be calculated for a non-probability sample.

N

NUMBERS and Measurement

What specifically will we measure, and at what level of precision?

Four levels of measurement

Nominal	Categories, no order	*Counting, mode, chi-square*	**Brand used most often; gender**
Ordinal	Ordered, unequal intervals	*Rank order, median, non-parametric*	**Satisfaction rank; education level**
Interval	Equal intervals, no true zero	*Mean, SD, t-test, ANOVA, regression*	**5–7 pt Likert satisfaction ratings**
Ratio	Equal intervals + true zero	*All arithmetic; full parametric analysis*	**Revenue; number of purchases**

Likert scales are technically ordinal, but conventionally treated as approximately interval – acknowledge the assumption in the blueprint rather than making it silently.

T

TECHNIQUES of Analysis

The final component – and the one most commonly absent from student proposals.

Pre-commit before data collection, not after

Descriptive	*Frequencies, means, cross-tabs*	Simply measuring & describing the sample
Bivariate	*Chi-square, t-test, ANOVA, correlation*	Testing if two variables relate or differ across groups
Multivariate	*Multiple/logistic regression, factor, cluster, conjoint*	Controlling for multiple variables, classifying, modelling complexity

***Skipping this invites HARKing* – Hypothesising After Results are Known**
Running many analyses and reporting only the interesting ones.
Pre-committing the plan is a commitment to rigour.

Common Mistakes

1. Treating ‘mixed methods’ as a complete Evidence Type specification – it says nothing about source, sequence, or integration.
2. Conflating unit of analysis with the respondent – the entity you conclude about isn’t always who you collected data from.
3. Writing a sample size with no justification – no link to population size, confidence, or the planned analysis.
4. Specifying analysis techniques (T) without checking consistency against measurement levels (N).
5. Treating BLUEPRNT as a checklist filled independently, rather than a system to cross-check end to end.